

Exercise of Supervised Learning: SVM Part 1

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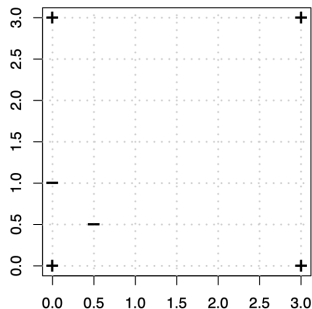
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Exercise 1: Soft Margin Classifier

The primal optimization problem for the two-class soft margin SVM classification is given by

$$\begin{aligned} \min_{\boldsymbol{\theta}, \theta_0, \zeta^{(i)}} \quad & \frac{1}{2} \|\boldsymbol{\theta}\|^2 + \sum_{i=1}^n \zeta^{(i)} \\ \text{s.t.:} \quad & y^{(i)}(\boldsymbol{\theta}^T \mathbf{x}^{(i)} + \theta_0) \geq 1 - \zeta^{(i)}, \\ & \zeta^{(i)} \geq 0, \quad \forall i = 1, \dots, n. \end{aligned}$$

(a) Add the decision boundary to the figure for $\hat{\boldsymbol{\theta}} = (1, 1)^T$, $\hat{\theta}_0 = -2$. (NB: This is the approximate optimum for $C = 10$).



Solution to Exercise 1 (a)

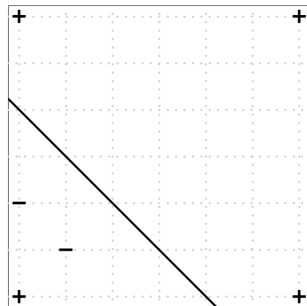
Hyperplane:

$$\theta_1 x_1^{(i)} + \theta_2 x_2^{(i)} + \theta_0 = 0$$

Plug in $\hat{\theta} = (1, 1)^T$, $\hat{\theta}_0 = -2$; solve for x_2 :

$$x_2 = -x_1 + 2$$

$\rightsquigarrow$ decision boundary.



Exercise 1 (b)

(b) Identify the points on the margin hyperplanes and compute the values of their slack variable $\zeta^{(i)}$.

Exercise 1 (b): derive $\zeta^{(i)}$ from $y^{(i)} f(\mathbf{x}^{(i)})$

Shorthand $m^{(i)} := y^{(i)}(\boldsymbol{\theta}^T \mathbf{x}^{(i)} + \theta_0) = y^{(i)} f(\mathbf{x}^{(i)})$ (geometric margin = $m^{(i)} / \|\boldsymbol{\theta}\|$).

s.t. $m^{(i)} \geq 1 - \zeta^{(i)}, \quad \zeta^{(i)} \geq 0.$

Minimizing $\sum_{i=1}^n \zeta^{(i)} \Rightarrow$ each $\zeta^{(i)}$ takes its smallest feasible value:

$$\zeta^{(i)} = \max(0, 1 - m^{(i)}) \quad (\text{hinge loss}).$$



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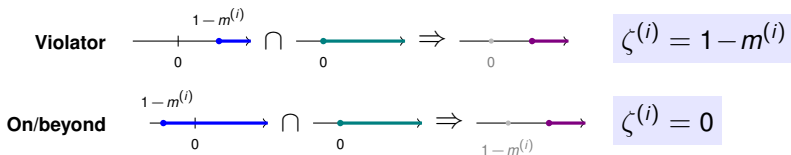
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Solution to Exercise 1 (b): point (0, 0)

At the optimum $\hat{\boldsymbol{\theta}} = (1, 1)^T$, $\hat{\theta}_0 = -2 \Rightarrow f(\mathbf{x}) = x_1 + x_2 - 2$, so $m^{(i)} = y^{(i)} (x_1^{(i)} + x_2^{(i)} - 2)$.

► $(0, 0)$, $y^{(1)} = +1$: $m^{(1)} = 1 (0 + 0 - 2) = -2$ (< 1 , violates margin)

► apply the recipe: $\zeta^{(1)} = \max(0, 1 - m^{(1)}) = \max(0, 3) = 3$

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Same recipe: $m^{(i)} = y^{(i)} (x_1^{(i)} + x_2^{(i)} - 2)$, $\zeta^{(i)} = \max(0, 1 - m^{(i)})$.

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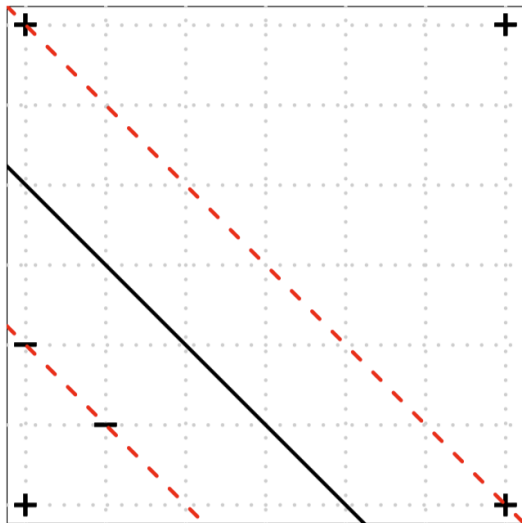
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Solution to Exercise 1 (b): Continued



Exercise 2: Optimization

Write your own stochastic subgradient descent routine to solve the soft-margin SVM in the primal formulation.

Hints:

- ▶ Use the regularized-empirical-risk-minimization formulation, i.e., an optimization criterion without constraints.
- ▶ No kernels, just a linear SVM.
- ▶ Compare your implementation with an existing implementation (e.g. `kernallab` in R. Are your results similar? Note that you might have to switch off the automatic data scaling in the already existing implementation.

Solution to Exercise 2: PEGASOS

Unconstrained regularized ERM with the hinge loss $\max(0, 1 - y^{(i)} f(\mathbf{x}^{(i)}))$ from (b).

SGD step: shrink θ (the penalty); if a point is inside the margin, add the hinge subgradient.

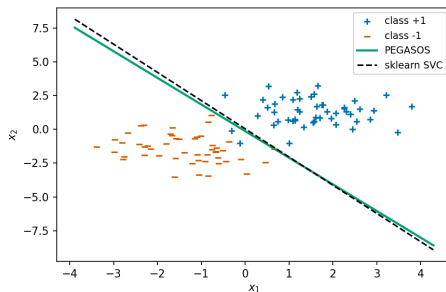
```
def pegasos_linear(y, X, nr_iter=50000, lam=1.0, alpha=0.01, seed=0):  
    # soft-margin SVM in the primal via stochastic subgradient descent  
    # X: design matrix with leading 1s column (intercept); y in {-1,+1}  
    rng = np.random.default_rng(seed)  
    n, p = X.shape  
    theta = rng.standard_normal(p)  
    for _ in range(nr_iter):  
        i = rng.integers(n)  
        f_i = X[i] @ theta # margin score  
        theta = (1 - lam * alpha) * theta # weight decay (L2 penalty)  
        if y[i] * f_i < 1: # inside margin: subgradient step  
            theta = theta + alpha * y[i] * X[i]  
    return theta
```

Solution to Exercise 2: check against sklearn

```
# two Gaussians (cf. mlbench.twonorm)
rng = np.random.default_rng(2)
a = 2 / np.sqrt(2)
Xp = rng.standard_normal((50,2)) + a
Xn = rng.standard_normal((50,2)) - a
X = np.vstack([Xp, Xn])
y = np.r_[np.ones(50), -np.ones(50)]

Xb = np.c_[np.ones(100), X]    # intercept
theta = pegasos_linear(y, Xb, lam=1/100)

# reference: sklearn linear SVM
svc = SVC(kernel="linear", C=1).fit(X, y)
```



Both misclassify 2/100; PEGASOS and sklearn boundaries nearly coincide.